Genes, Carbon Dioxide and Adaptation.

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Examining the dogmatic models of life and the world in which life exists, we can better understand the nature of the existing barriers to constructive work. The Central Dogma of the molecular geneticists, in their own words, was that information flows only from DNA to RNA, and from RNA to protein, never in the other direction. The Central Dogma was formulated to suppress forever the Lamarckian idea of the inheritance of acquired characters, that Weismann's amputation of the tails of a multitude of mice had attempted to deal with earlier in the history of genetics. The Central Dogma continues to be influential, even after a series of revisions. Until the 1990s, the only "practical" fruit of genetics had been genocide, but now it has become possible to insert genes into bacteria, and to use the bacteria to produce industrial quantities of specific proteins. In principle, that could be useful, although bovine growth hormone poses a threat to the health of both people and cows, human growth hormone poses a threat to athletes and old people, and human insulin could increase the number of treated diabetics. A deranged culture will put anything cheap to bad use. The ability to make organisms produce foreign proteins confirms that information can flow from DNA to protein, but as that technology was being developed, the discovery of retroviruses showed that the Central Dogma of molecular genetics was wrong, RNA is a very significant template for the production of DNA. And the scrapie prion shows that proteins can be infectious, passing along information without nucleic acids as the agent of transmission. The directed mutations demonstrated by John Cairns and others have thoroughly destroyed the Central Dogma of molecular genetics, even as it applied to the simplest organisms, but molecular genetics survives as an industrial and forensic technology. Although evidence suggests that about 2% of human diseases involve the inheritance of an abnormal protein, the exact way the disease develops is never as clear as the geneticists would imply. And the major diseases, cancer, diabetes, heart disease, Alzheimer's, epilepsy, depression, etc., that are so often blamed on "genes," are so poorly understood that it is arbitrary and crazy to talk about the way genes "cause" them. People who had never had a problem with diabetes in their culture, very soon suffered from the same rate of diabetes as their neighbors when they immigrated into Israel and began eating the European style diet. The interesting thing about the genetic explanation for disease is how its proponents can believe what they are saying. If you read Konrad Lorenz's writings on racial hygiene, you can imagine that he might have really come to believe what he was saying, even if it was an invention that earned him personal prestige and revenge against people who were reluctant to accept his ideas of cultural excellence and inferiority. When I listened to Gunther Stent praising the doctrine he had taken straight from Konrad Lorenz's original genocide papers, I wondered how a German who had escaped the holocaust with his Jewish family when he was nine years old could talk about those doctrines without anger, and without pointing out the purpose for which they had been created. In the audience, a professor who had been a refugee from Hungary defended the doctrine, saying that a man and his work have nothing to do with each other, though the whole content of the doctrine was that a man and his work are identical, because his behavior is determined by his genes. These were mature, internationally known intellectuals, who made the most amazingly self-contradictory statements without embarrassment, because they were committed, for some deep, mysterious reason, to the doctrine of genetic determinism. If these refugees could espouse the rationale for "racial hygiene" as their own, I suppose it isn't so hard to understand that people can devote their life to studying the genetics of diabetes, even though diabetes has appeared suddenly in one generation of immigrants when their diet was suddenly changed, a massive fact that bluntly contradicts the genetic doctrine.

There is something very deep in our culture that loves genetics. One of the cultural trends that makes genetic determinism attractive is the theory of radical individualism, something that has grown up with protestant christianity, according to some historians. Roger Williams' work in nutrition seemed to be powered by this idea of individual genetic uniqueness, and in his case, the idea led him to some useful insights--he suggested that the environment could be adjusted to suit the highly specific needs of the individual. This idea led to the widespread belief that nutritional supplements might be needed by a large part of the population. Extreme nurturing of the deviant individual is the opposite extreme from the Lorenzian-Hitlerian solution, of eliminating everyone who wasn't a perfect Aryan specimen. But Williams' genetic doctrine assumed that our nutritional needs were primarily inborn, determined by our unique genes. However, there is a famous experiment in which rats were made deficient in riboflavin, and when their corneal tissue showed evidence of the vitamin deficiency, they were given a standard diet. However, the standard diet no longer met the needs of their eye tissue, and during the remainder of the observation period, only a dose of riboflavin several times higher than normal would prevent the signs of deficiency. A developmental change had taken place in the cornea, making its vitamin B2 requirement abnormally high. If we accept the epigenetic, developmental idea of metabolic requirements, our idea of nurturing environmental support would consider the long-range effects of environmental adequacy, and would consider that much disease could be prevented by prenatal support, and by avoiding extreme deficiencies at any time. Williams himself emphasized the importance of prenatal nutrition in disease prevention, so he wasn't a genetic totalitarian; combining the idea of unique genetic individuality with the recognition that malnutrition causes disease, led him to believe in the necessity for nutitional adequacy, rather than to the extermination of the sick, weak, or different individuals. The idea of "genetic determinism" says that our traits are the result of the specific proteins that are produced by our specific genes. The doctrine allows for some gradations, such as "half a dose" of a trait, but in practice it becomes a purely subjective accounting for everything in terms of mysterious degrees of "penetrance" of genes, and interactions with unknown factors. Proteins, that supposedly express our genetic constitution, include enzymes, structural proteins, antibodies, and a variety of protein hormones and peptide regulatory molecules. Every protein, including the smallest peptide (except certain cyclic peptides), contains at least one amine group, and usually several. Amine groups react spontaneously with carbon dioxide, to form carbamino groups, and they can also react, nonenzymically, with sugars, in the reaction called glycation or glycosylation. These chemical changes alter the functions of the proteins, so that hormones and their "receptors," tubules and filaments, enzymes and synthetic systems, all behave differently under their influences. (The proteins' electrical charge, relationship to water and fats, and shape, change quickly and reversibly as the concentration of carbon dioxide changes; in the absence of carbon dioxide, these properties tend to change irreversibly under the influence of metabolic stress.) This is the clearest, and the most powerful, instance of metabolic influence on biological structure. That makes it very remarkable that it has been the subject of so few publications. I think the absence of discussion of this fundamental biological principle can be understood only in relation to the great importance it has for a new understanding of development and inheritance--it is an easily documented process that will invalidate some of the most deeply held beliefs of most of the people who are influential in science and politics. I will continue discussing some of these implications in newsletters on imprinting, degenerative diseases, heart attacks, high blood pressure, and other special biological questions, but I think the most important work that remains to be done is to work out the exact mechanisms by which metabolic energy, expressed largely by factors such as the ratio of carbon dioxide to lactic acid, guides both development and evolution. These ideas will have to take into account the actual resources of the world, as well as the internal processes and resources of the organism. Each development in the organism, whether it leads to maturation or to degeneration, consists of responses to and interactions with specific environments. Curiosity, esthetics, creativity, and stimulation are necessarily and deeply linked to metabolic efficiency and structural-anatomical development. For example, the known effects of stimulation and success (or isolation and depression) on brain anatomy and function should be linked meaningfully with metabolic, hormonal and dietary processes. There is a large amount of information available that could be put to practical use, but there are still important ideological barriers to be overcome. Marshalling the information needed to optimize our own development runs counter to the program of our technical-scientific culture, which prefers to believe that degeneration is programmed, while emergent evolution is unforeseeable. But, if an optimization project is presented as a way to forestall the "programmed degeneration," it might succeed in becoming part of the culture. Vernadsky's idea of the Noosphere differs from the Gaia hypothesis (that the world is a self-regulating organism-like system) in the intrinsic directionality of Vernadsky's Noosphere, which makes the course of human society crucial for the fate of the planet. It proposes that planets, like organisms, are going somewhere. The Gaia hypothesis is increasingly being interpreted as a justification for feeling no responsibility for the effects of technology on the environment, and some people are expressing that view of the world as essentially a justification for any vandalism that may come along. Kary Mullis, for example, says that mass extinctions of organisms have occurred in the past, and so it's just natural for species to become extinct, and it isn't appropriate to be concerned about the extinctions that are being caused by civilization's technological depredations. In the Noosphere, global warming and increased carbon dioxide would represent an advance toward a higher state of "metabolism" of the world, and this would support the emergence of new biological forms from those existing. But if whole systems of life are destroyed before that happens, the biological achievements of the past could be lost irretrievably; there is no guarantee that the system will continue to work, if major sectors are deleted from the interacting systems. Even in terms of the Gaia conception, that the earth is like an organism, consider what the loss of genetic complexity means for an organism. Sometimes, for example, things that happen to an individual lead to sterility several generations later, although the procedure didn't seem lethal for the individual or its immediate descendants. The whole idea of "evolution" is that the past is preserved within the present, or that the present is built upon the accomplishments of the past. The idea that evolution has been "random," and that the world is simply self-regulating, might seem liberating to those who hate the idea that they might be intrinsically responsible for anything outside of themselves, but it is liberating only in the way that a vandal's manifesto might be, declaring the world to be their playground. The problem with such a manifesto of irresponsibility is simply that it is built upon the same system of cultural assumptions that produced Nazi eugenics, and that those assumptions are false. The political assumptions of the people who controlled scientific institutions were built into a set of pseudo-scientific doctrines, which continue to be valued for their political and philosophical implications. For hundreds or thousands of years, the therapeutic value of carbonated mineral springs has been known. The belief that it was the water's lively gas content that made it therapeutic led Joseph Priestley to investigate ways to make artificially carbonated water, and in the process he discovered oxygen. Carbonated water had its medical vogue in the 19th century, but the modern medical establishment has chosen to define itself in a way that glorifies "dangerous," "powerful" treatments, and ridicules "natural" and mild approaches. The motivation is obvious--to maintain a monopoly, there must be some reason to exclude the general public from "the practice of medicine." 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But if whole systems of life are destroyed before that happens, the biological achievements of the past could be lost irretrievably; there is no guarantee that the system will continue to work, if major sectors are deleted from the interacting systems. Even in terms of the Gaia conception, that the earth is like an organism, consider what the loss of genetic complexity means for an organism. Sometimes, for example, things that happen to an individual lead to sterility several generations later, although the procedure didn't seem lethal for the individual or its immediate descendants. The whole idea of "evolution" is that the past is preserved within the present, or that the present is built upon the accomplishments of the past. 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The Central Dogma of the molecular geneticists, in their own words, was that information flows only from DNA to RNA, and from RNA to protein, never in the other direction. The Central Dogma was formulated to suppress forever the Lamarckian idea of the inheritance of acquired characters, that Weismann's amputation of the tails of a multitude of mice had attempted to deal with earlier in the history of genetics. The Central Dogma continues to be influential, even after a series of revisions. Until the 1990s, the only "practical" fruit of genetics had been genocide, but now it has become possible to insert genes into bacteria, and to use the bacteria to produce industrial quantities of specific proteins. In principle, that could be useful, although bovine growth hormone poses a threat to the health of both people and cows, human growth hormone poses a threat to athletes and old people, and human insulin could increase the number of treated diabetics. 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One of the cultural trends that makes genetic determinism attractive is the theory of radical individualism, something that has grown up with protestant christianity, according to some historians. Roger Williams' work in nutrition seemed to be powered by this idea of individual genetic uniqueness, and in his case, the idea led him to some useful insights--he suggested that the environment could be adjusted to suit the highly specific needs of the individual. This idea led to the widespread belief that nutritional supplements might be needed by a large part of the population. Extreme nurturing of the deviant individual is the opposite extreme from the Lorenzian-Hitlerian solution, of eliminating everyone who wasn't a perfect Aryan specimen. But Williams' genetic doctrine assumed that our nutritional needs were primarily inborn, determined by our unique genes. However, there is a famous experiment in which rats were made deficient in riboflavin, and when their corneal tissue showed evidence of the vitamin deficiency, they were given a standard diet. However, the standard diet no longer met the needs of their eye tissue, and during the remainder of the observation period, only a dose of riboflavin several times higher than normal would prevent the signs of deficiency. A developmental change had taken place in the cornea, making its vitamin B2 requirement abnormally high. If we accept the epigenetic, developmental idea of metabolic requirements, our idea of nurturing environmental support would consider the long-range effects of environmental adequacy, and would consider that much disease could be prevented by prenatal support, and by avoiding extreme deficiencies at any time. Williams himself emphasized the importance of prenatal nutrition in disease prevention, so he wasn't a genetic totalitarian; combining the idea of unique genetic individuality with the recognition that malnutrition causes disease, led him to believe in the necessity for nutitional adequacy, rather than to the extermination of the sick, weak, or different individuals. The idea of "genetic determinism" says that our traits are the result of the specific proteins that are produced by our specific genes. The doctrine allows for some gradations, such as "half a dose" of a trait, but in practice it becomes a purely subjective accounting for everything in terms of mysterious degrees of "penetrance" of genes, and interactions with unknown factors. Proteins, that supposedly express our genetic constitution, include enzymes, structural proteins, antibodies, and a variety of protein hormones and peptide regulatory molecules. 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I will continue discussing some of these implications in newsletters on imprinting, degenerative diseases, heart attacks, high blood pressure, and other special biological questions, but I think the most important work that remains to be done is to work out the exact mechanisms by which metabolic energy, expressed largely by factors such as the ratio of carbon dioxide to lactic acid, guides both development and evolution. These ideas will have to take into account the actual resources of the world, as well as the internal processes and resources of the organism. Each development in the organism, whether it leads to maturation or to degeneration, consists of responses to and interactions with specific environments. Curiosity, esthetics, creativity, and stimulation are necessarily and deeply linked to metabolic efficiency and structural-anatomical development. For example, the known effects of stimulation and success (or isolation and depression) on brain anatomy and function should be linked meaningfully with metabolic, hormonal and dietary processes. There is a large amount of information available that could be put to practical use, but there are still important ideological barriers to be overcome. Marshalling the information needed to optimize our own development runs counter to the program of our technical-scientific culture, which prefers to believe that degeneration is programmed, while emergent evolution is unforeseeable. But, if an optimization project is presented as a way to forestall the "programmed degeneration," it might succeed in becoming part of the culture. Vernadsky's idea of the Noosphere differs from the Gaia hypothesis (that the world is a self-regulating organism-like system) in the intrinsic directionality of Vernadsky's Noosphere, which makes the course of human society crucial for the fate of the planet. It proposes that planets, like organisms, are going somewhere. The Gaia hypothesis is increasingly being interpreted as a justification for feeling no responsibility for the effects of technology on the environment, and some people are expressing that view of the world as essentially a justification for any vandalism that may come along. Kary Mullis, for example, says that mass extinctions of organisms have occurred in the past, and so it's just natural for species to become extinct, and it isn't appropriate to be concerned about the extinctions that are being caused by civilization's technological depredations. In the Noosphere, global warming and increased carbon dioxide would represent an advance toward a higher state of "metabolism" of the world, and this would support the emergence of new biological forms from those existing. But if whole systems of life are destroyed before that happens, the biological achievements of the past could be lost irretrievably; there is no guarantee that the system will continue to work, if major sectors are deleted from the interacting systems. Even in terms of the

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